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1. a. B is optimal. Because B is on the budget line and is the tangency point of Indifference curve

b. $Y = P_a C_a^* + P_b C_b^*$

$$\frac{P_a}{P_b} = \frac{\alpha u'(C_a^*)}{(1-\alpha)u'(C_b^*)}$$

c. $$\begin{cases} C_b^* = \dfrac{Y - C_a^* P_a}{P_b} \\ \dfrac{P_a}{P_b} = \dfrac{\alpha C_b^*}{(1-\alpha) C_a^*} \end{cases} \therefore \begin{cases} C_a^* = \dfrac{\alpha Y}{P_a} \\ C_b^* = \dfrac{(1-\alpha)Y}{P_b} \end{cases}$$

2. a. $$\begin{cases} Y \geq C_0 + S \\ S \geq \dfrac{C_1}{1+r} \end{cases} \therefore Y \geq C_0 + \frac{C_1}{1+r}$$

b. $U = \ln(C_0) + \beta \ln[(Y - C_0)(1+r)]$, $C_1 = (Y - C_0)(1+r)$

$u' = \dfrac{1}{C_0} - \dfrac{\beta}{Y - C_0} = 0$ $\therefore$ $$\begin{cases} C_0 = \dfrac{Y}{1+\beta} \\ C_1 = \dfrac{\beta}{1+\beta} Y(1+r) \end{cases}$$

C. $\dfrac{C_1^*}{C_0^*} = \beta(1+r)$ $\therefore$ when β or r increases, $\dfrac{C_1^*}{C_0^*}$ increases.

3. $\because v\left(\frac{C_1+C_2}{2}, \frac{C_1+C_2}{2}\right) = 2u\left(\frac{C_1+C_2}{2}\right)$

$v(C_1, C_2) = u(C_1) + u(C_2)$

$\therefore$ To show $v\left(\frac{C_1+C_2}{2}, \frac{C_1+C_2}{2}\right) > v(C_1, C_2)$ is the same as showing:

$$u\left(\frac{C_1+C_2}{2}\right) > \frac{u(C_1)+u(C_2)}{2}$$

Is just to show $u(C)$ is convex

$\therefore u'(C) = C^{-\nu} > 0$ $(C>0, \nu>0)$

$\therefore u''(C) = -\nu C^{-\nu-1} < 0$ $\therefore u(C)$ is convex, consumer will always prefer a smooth consumption.